PURPOSE:

Consistency in developing loss estimates is important to GRC clients. This can be difficult to achieve when different consultants with different perspectives are developing loss estimates at similar client facilities. A GRC report represents a “snapshot” of a facility at the time it is surveyed. Our clients want all the pictures taken with the same camera. This practice aims to clarify and standardize the process in which potential loss is estimated based on observed and projected (after recommendations are completed) conditions to better achieve the consistency desired by our clients.

This Fire Practice provides guidelines for developing loss estimates related to the peril of Fire. As such, the APL, PML and MFL should be based on a fire event occurring, even where Nat Haz loss estimates exceed the fire loss estimates.

Third party fire exposures should be considered, including adjacent buildings or tenants within the same building. Consider a multi-tenant warehouse/manufacturing facility where all operations, hazards and protection are not known. Without adequate separation, a fire starting in a tenant space could easily spread to client occupied space. Apply PML guidance for minimum distances between buildings or fire rated separation requirements, when evaluating the APL scenario.

Some client ASIs may require additional loss estimates (e.g., windstorm, flood, etc.) be provided in the report. This Fire Practice does not cover loss estimate calculations for other perils. Please refer to the applicable Fire Practice when other loss estimates are required.

For special occupancies or unique building construction features, please reference the applicable Fire Practice (e.g., FP5601 – Electronic Data Processing, etc.) for guidance on calculating loss estimates.

Contents

GENERAL 3
Legacy System Design Analysis....4
Differences in Standards 4
Business Interruption 5
Specific Occupancies and Hazards:....5
ANTICIPATED PROBABLE LOSS (APL)....6
Rules of Thumb for APL Loss Estimates:....6
Machinery and Equipment Fires 7
Flammable and Combustible Liquids 8
Combustible Dust....8
Figure 1A – Typical APL Scenarios for Storage Occupancies 9
Figure 1B – Typical APL Scenarios for Non-Storage Occupancies (Non-Sensitive) [1]....10
Figure 1C – Typical APL Scenarios for Non-Storage Occupancies (Sensitive) [1]....11
Table 1A – Typical APL Loss Estimate Damage 12
Table 1B – Typical APL Loss Estimate Damage – Sprinklers with Deficiencies....14
PROBABLE MAXIMUM LOSS (PML)....16
Rules of Thumb for PML Loss Estimates:....16
Special Considerations for High Rise Buildings: 17
Figure 2 – High Rise PML Scenarios 18
Table 2 – PML High Rise Damage 18
MAXIMUM FORESEEABLE LOSS (MFL) 19
Rules of Thumb for MFL Loss Estimates (non High Rise): 19
Table 3a – Minimum MFL Space Separation (ft) 19
Table 3b – Minimum MFL Space Separation (meters)....19
Determining MFL Damage:....20
Table 4 – Typical MFL Loss Estimate Damage Percentages 20
Determining MFL Area for High Rise Buildings:....24

GENERAL

Loss estimate guidelines are not necessarily related to protection standards. A client's protection guidelines may state that sprinklers are not to be provided in computer or electrical equipment rooms. Under such conditions, GRC consultants should not recommend sprinkler protection in those areas for that client. However, the fact that sprinklers are not recommended does not affect the loss estimates. The potential loss in an unsprinklered computer or electrical room is the same, regardless of whether or not sprinklers are recommended.

Probability is a significant factor in risk management decisions. In general, standard property loss control industry practices appear to ignore probabilities in preparing loss estimates. However, there are implied probabilities in loss estimate definitions like Anticipated Probable Loss, Probable Maximum Loss, and Maximum Foreseeable Loss. The majority of fires and industrial mishaps do not result in significant loss. That is because, in most cases, fire protection equipment, operator training, emergency response procedures, etc., work as designed to limit the loss.

Loss estimation is far from an exact science. It is, in reality, an effort to put an upper limit on the potential loss under defined conditions or probability range. As such, loss estimates normally should be limited number to two significant digits. In no case should loss estimates imply an accuracy of better than US$100,000 (unless the total LE is less than $100,000, or the LE is equal to 100%). Representations of greater accuracy cannot normally be justified. They indicate the writer can read the display of a calculator but does not understand the meaning of the figure and has not checked to see if it makes sense.

This document provides a baseline for calculating loss estimates but does not consider all potential circumstances. Please reference the numbered footnotes following the applicable tables for adjustments based on local conditions. Engineering judgement should be exercised when loss estimates are justifiably higher or less than those in the tables below.

Account service instructions must also be considered when drafting loss estimates. Consult with the account engineer when deviating from this standard or the ASI.

It is important to recognize that the loss scenario is just as important as the value of the loss estimate itself. The loss scenario should be well explained, such that the reader can clearly follow the scenario, and understand how you arrived at your conclusion. The same applies to the Loss Estimate value calculation. This is not only important for the reader to understand but might become equally important if a loss occurs and we need to explain how we arrived at the values identified in this document. Consider the following 10 steps to help with crafting your loss scenario:

- Values
    1. What are the values on site?
    2. How are the values distributed?

- Location of fire event
    3. Where to locate the fire?
    4. When to initiate the fire?

- Fire scenario
    5. How does the fire start?[1]
    6. How does the fire develop and spread?

7. What stops the fire?

- Outcome
  8. What is the damage to assets?
  9. How is the business interrupted?

- Recovery
  10. How are operations restored?

Notes:
1. Fire loss scenarios need not contemplate a specific ignition source but recognize that a fire will occur.

Legacy System Design Analysis

Existing systems that were designed and installed in accordance with previous editions of the applicable standard (NFPA, FMDS, VdS, etc.) must be thoroughly evaluated by the field engineer. Where the analysis reveals that protection is inadequate based on current standards or test data, the account team should be notified. The account team should consult with the client regarding these changes and develop an account specific plan for addressing recommendations and loss estimates, subject to the comments below:

Deficiencies that result in an APL=PML scenario, must be considered inadequate. The ASI may be updated to indicate recommendations are not required for these deficiencies. Loss estimates must always be updated to reflect the identified deficiency. The legacy design must also be addressed as *Inadequate* in the report.

When APL≠PML, legacy designs may be considered tolerable (not adequate) when developing loss estimates. In this case, legacy designs must be addressed as *Tolerable* in the report. In the APL flowcharts, assume the system meets 80% of the base-of-riser flow requirements. Other sprinkler deficiencies will require Table 1B to be examined, but the legacy design will not.

For all sprinkler systems without hydraulic design information (including pipe schedule systems), every effort must be made to determine the adequacy of the system. This can include procurement of existing as-built drawings or sketching the system and performing a hydraulic analysis (see FP4000). Making a recommendation for the client to obtain an as-built drawing is unacceptable except in the rare occasions where critical sections of the sprinkler system piping are hidden or inaccessible.

For all designs that are not addressed in recognized standards (NFPA, FMDS, VdS, etc), contact the account engineering team and the chief engineer to discuss the design and loss estimates associated.

Differences in Standards

GRC **applies** NFPA standards for acceptable design criteria. A gap analysis should be performed where other standards are used (Country specific or client requirement). Where design criteria listed in these standards is inferior to NFPA, then NFPA design criteria should be applied. For protection requirements outside the scope of NFPA, other standards may be used.

Where fire standards are updated in response to tests completed by an accredited body, the most current updates to accepted fire standards take precedence over legacy or previously accepted designs.

Field engineers must identify deficiencies reflecting the newest fire standards and issue pertinent recommendations. Loss estimates must be updated to reflect the newly identified deficiencies.

Where clients adopt other standards, the most current updates to accepted fire standards take precedence.

An example includes the most recent tests completed by UL for K-14 ESFR sprinkler heads and clearance requirements.

Business Interruption

A loss scenario may occur in the area with the most potential property damage **AND/OR** in the area with the most potential business interruption, whichever results in the largest monetary value loss.

Areas with the least redundancy and most impact to the business operations should be considered for loss scenarios. The occupancy hazard should drive the loss and business interruption resulting from the loss.

The business interruption calculations provided throughout this document are estimates and indicate the minimum expected interruption for the associated occupancy/construction. Expected lead times for equipment must be considered when generating business interruption estimates. Plant personnel must be consulted regarding lead/repair times on their equipment and the impact it would have on production.

Specific Occupancies and Hazards:

Where occupancy or hazard specific fire practices exist, they take precedence over this fire practice and should be consulted for developing loss estimates. Examples include:

Data Centers and Electronic Equipment Rooms (White Space):

See FP 5061

Exterior Insulation Finishing Systems:

See FP 5910

Insulated Metal Panels:

See FP 5915

OxyReduct Systems:

See FP 8511

ANTICIPATED PROBABLE LOSS (APL)

GRC's definition of the APL is the maximum likely loss that would result with all protection systems functioning **as installed**, employees responding in accordance with internal protocols/procedures per training received, and outside agencies responding per standard procedures. This definition is reflected in the wording of the Loss Estimates section of the standard report template. In probability terms, this definition implies that, should a loss occur, its consequences will be less than the APL estimate 70-80% of the time. Where existing fire protection features and controls are inadequate to protect the hazard, an uncontrolled fire in the area could occur. This may result in an APL = PML scenario for a fire occurring in a **PARTICULAR** area. The APL need not be in the same fire area as the PML or MFL scenario.

Nearly 60% of sprinkler system failures are due to the systems being shut off. The same methodology should be applied for areas with impaired fire protection, especially where protection is not expected to be restored quickly. Where impairments are identified onsite (emergency or preplanned), carefully review and document the established procedures and expected timeline. Even minor system component replacements can take weeks. Major system repairs (e.g., frozen sprinklers in refrigerated warehouses), or fire pump replacements may take months.

APL-type loss estimates should be the primary basis for evaluating existing conditions. In other words, the cost-effectiveness of recommendations should be based on the difference between the current APL estimate and the APL estimate with the recommendation completed.

Important Note: *The APL is a localized event that is a function of occupancy and the fire protection provided. Additions to the site, outside of the APL area, do not typically affect the APL. Inventory value changes* ***outside*** *of the APL area do not typically affect the APL. Therefore, APL loss scenarios that state APL = 1% (or any other percent) of reported property values are normally not acceptable. APL loss scenarios should be presented as monetary values, unless specific guidance is provided in the ASI, report template, or in the case where monetary values are not provided.*

Significant changes at the facility (including removing or adding buildings, equipment, and/or stock) can require the APL scenario to be changed.

Rules of Thumb for APL Loss Estimates:

Use the Figure 1 flow charts as a guideline for developing APL estimates for typical occupancies (Figure 1A for storage and Figure 1B for non-sensitive, non-storage and Figure 1C for sensitive, non-storage). The flow charts and tables assume the following:

- For light and ordinary hazard occupancies with adequately designed (per NFPA or FM) sprinkler systems, seven sprinkler heads will operate.
- For extra hazard occupancies (no flammable/combustible liquids) with adequately designed (per NFPA or FM) standard sprinkler systems, 12 sprinkler heads will operate. This also applies to rack storage areas in which case half of operating sprinklers will be in-racks and half at ceiling.
- Sprinkler systems that can provide at least 80% of the design density over the design area (without hose streams), and where significant flammable/combustible liquids or dust are not present, will control a fire by operating all sprinkler heads in the design area.
- For rack storage areas with inadequate protection because of:
  - Adequate in-racks, but weak ceiling (< 100% but >80% density provided)

– or –

- Adequate ceiling, but inadequate in-racks
  - Must meet layout and spacing requirement in NFPA 13
  - > 80% of flow or pressure provided, with required number of heads operating

– and –

- No flammable or combustible liquids
- No aerosols
- No open top, solid five-sided containers

all design area sprinklers will operate to control the fire.

- For multi-story buildings, water damage should be considered for areas directly below the sprinkler operating area. Water damage could be expected on multiple floors (typically 2 – 3) directly below the fire floor, each with a lesser degree. The same concept applies to smoke damage for floors above. Smoke damage will be experienced throughout the room(s) in which sprinklers operate with additional (lesser) smoke damage in the rooms directly adjacent to and above the sprinkler operating area (typically 2 – 3 floors). This is especially true for high rise structures.

  Where high rise structures possess combustible construction or combustible exterior cladding (i.e. EIFS, IMP panels, or other plastic construction materials). If there are significant sprinkler issues or significant external fire exposure, APL = PML.

- Long term impairments (greater than one month estimated restoration time) must be considered when writing an APL loss scenario as if the corresponding system is not installed in the following roadmaps. Appropriate recommendations should be considered.

Machinery and Equipment Fires

Fires involving machinery and equipment can result in significantly higher PD and BI losses and should be thoroughly evaluated on a case-by-case basis. This Fire Practice does not address every occupancy or unique situation where fires involve machinery and equipment. Table 1A below provides percent damage for equipment over the fire, water and smoke areas in general based on the occupancy and adequacy of protection. Fires involving machinery and equipment where flammable/combustible liquids are used, or combustible dust is produced, may result in significantly more damage to the equipment. This can also result in higher BI losses, especially where production bottlenecks exist.

For example, consider a hydraulic spray fire on a large metal stamping press or plastic injection molding machine. Even with appropriate sprinklers and interlocks, a large fire could still occur, resulting in a significant or even complete loss of a single piece of equipment. Sprinklers and interlocks will minimize fire damage to adjacent areas and equipment.

Flammable and Combustible Liquids

Facilities with flammable and combustible liquids need to be evaluated based on the quantity of liquid present, drainage and containment provisions, and other fire protection systems that can be used to minimize fire spread. Significant quantities, as defined below (per fire area), must be evaluated based on the adequacy of protection provided and other mitigating factors.

- Class I: 10 gallons
- Class II: 55 gallons
- Class III: 275 gallons
- Water-miscible: 55 gallons

Where significant quantities of flammable or combustible liquids are present all other applicable fire practices will apply and take priority over this standard (FP5300 is currently in development for flammable and combustible liquid loss estimate evaluation).

Current guidance for areas with significant flammable and/or combustible liquids or aerosol storage, and adequately designed protection systems, assume all sprinklers in the designed operating area will operate.

Combustible Dust

Facilities with combustible dust exposure must consider a combustible dust explosion as a potential APL scenario when deficiencies in passive/active protection features or loss prevention programs are identified.

A separate standard will be developed for in-depth analysis of combustible dust occupancy loss scenarios. FP5400 will be in development.

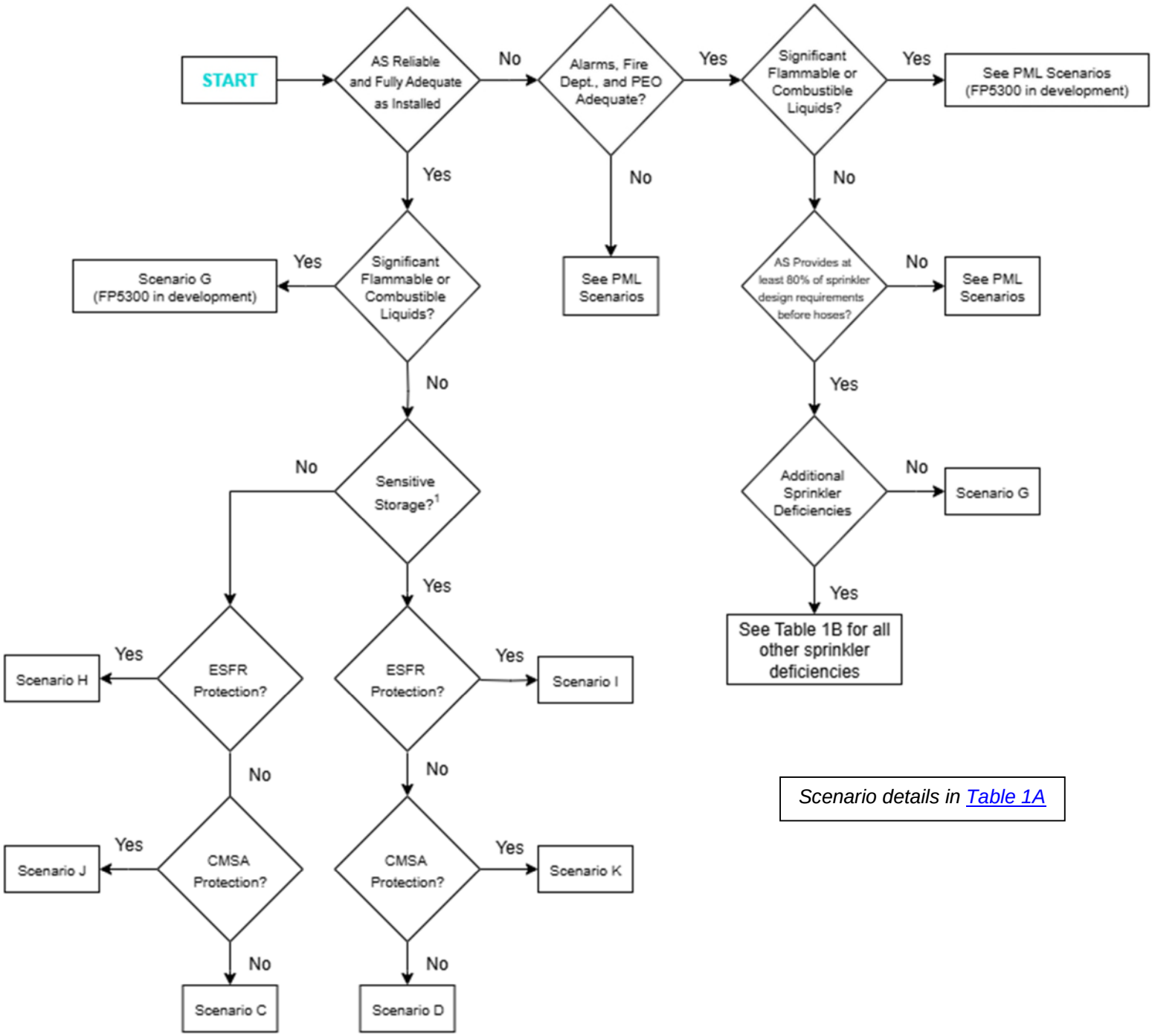


Figure 1A – Typical APL Scenarios for Storage Occupancies

Note:
1: Sensitive storage would include but are not limited to the following – Consumer Goods (Food, Beverage, Clothing), Sensitive Electronics, Pharmaceutical, and Medical Devices/Equipment.

Packaging supplies used for any of the above occupancies should also be considered.

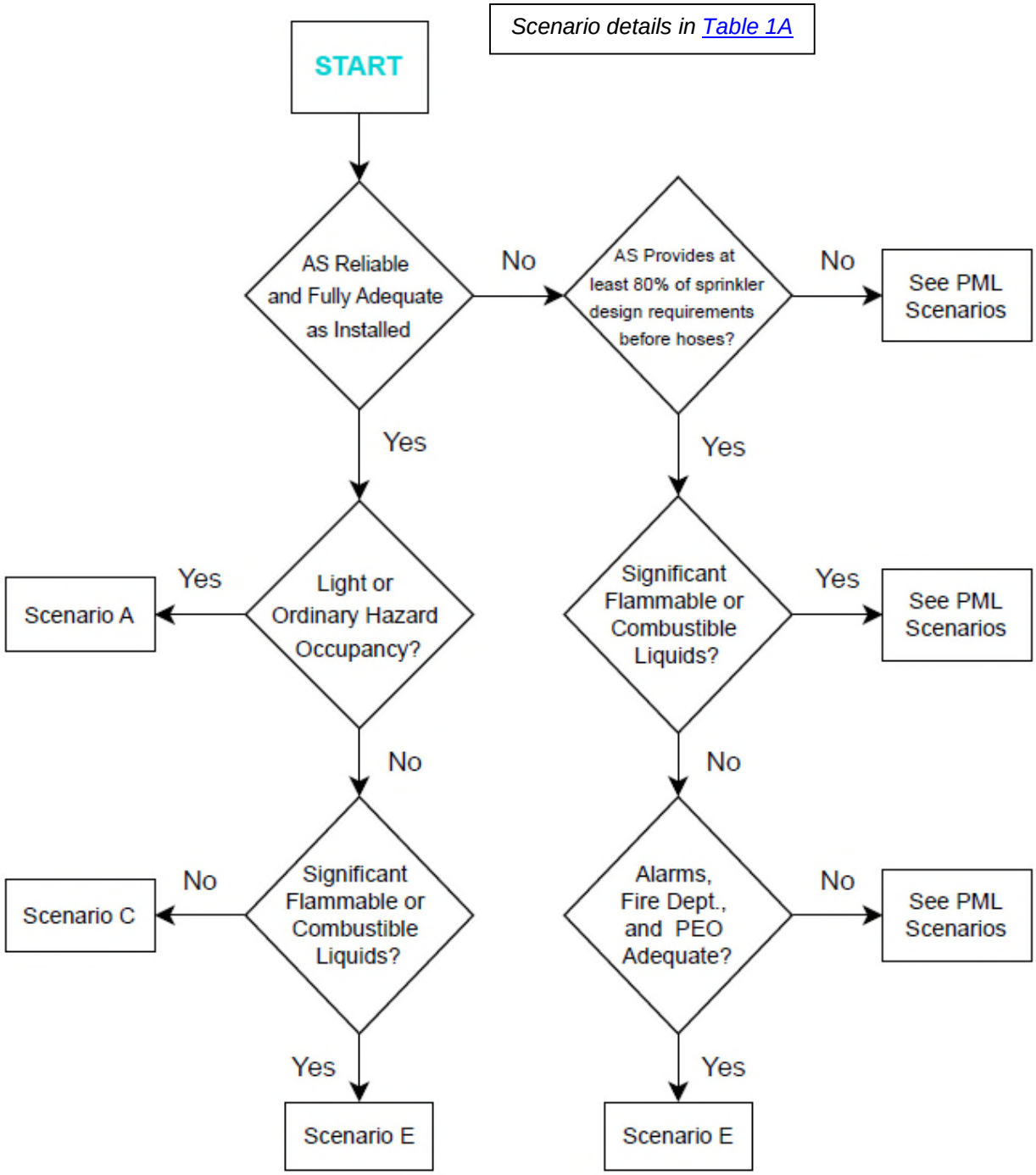


Figure 1B – Typical APL Scenarios for Non-Storage Occupancies (Non-Sensitive) [1]

Note:
1: Sensitive occupancies would include but are not limited to the following – Manufacturing of Consumer Goods (Food, Beverage, Clothing), Sensitive Electronics, Pharmaceutical, and Medical Devices/Equipment.

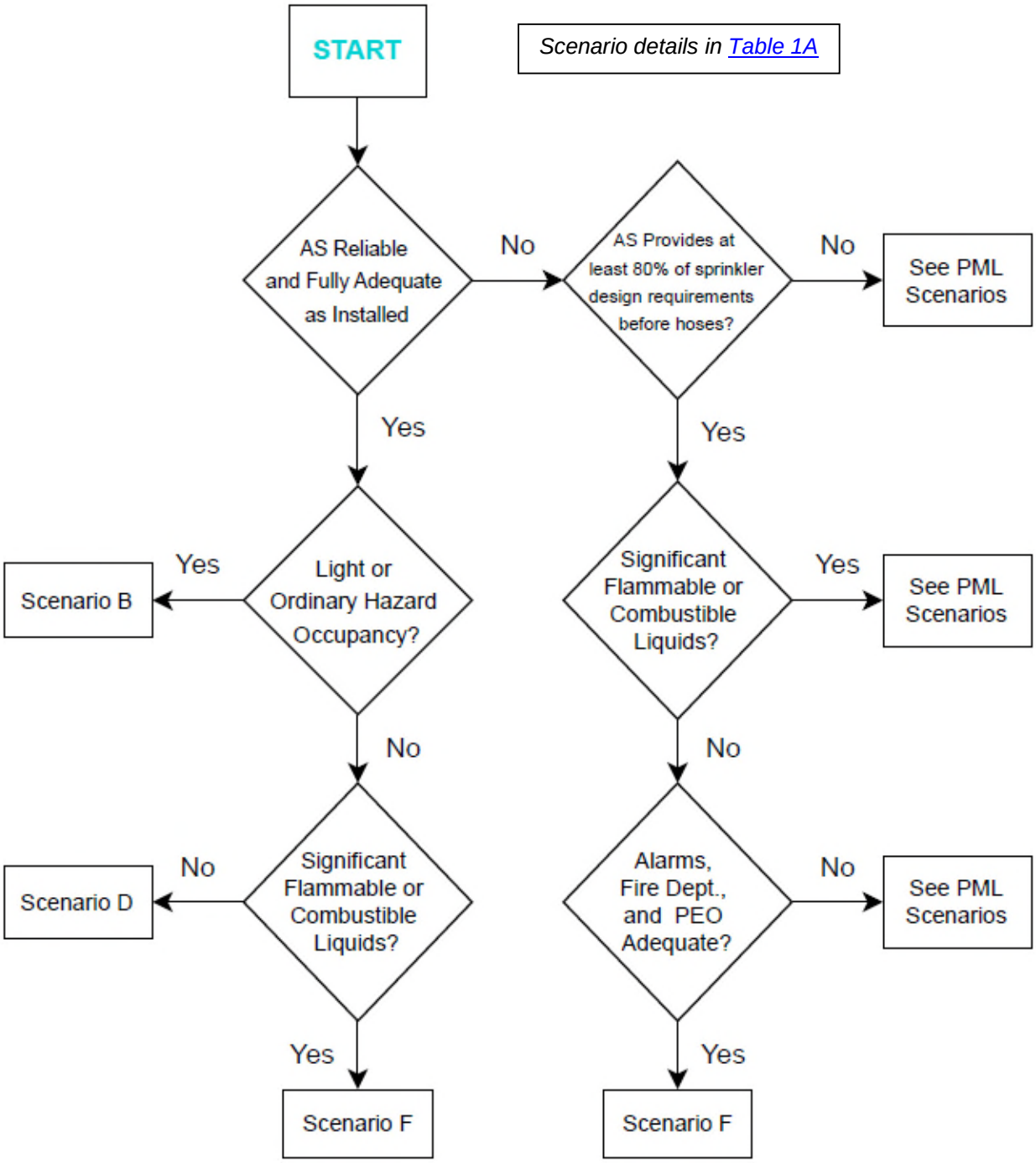


Figure 1C – Typical APL Scenarios for Non-Storage Occupancies (Sensitive) [1]

Note:
1: Sensitive occupancies would include but are not limited to the following – Manufacturing of Consumer Goods (Food, Beverage, Clothing), Sensitive Electronics, Pharmaceutical, and Medical Devices/Equipment.

Table 1A – Typical APL Loss Estimate Damage

Scenario	Damage From	Area Involved Wet System Type[1]	Area Involved Dry System Type[1]	Building Damage	Equipment Damage	Inventory /WIP Damage	BI
A	Fire	700 ft^2 (65 m^2)	1000 ft^2 (93 m^2)	25%	100%	100%	1 - 7 Days
	Water	1500 ft^2 (140 m^2)	2000 ft^2 (185 m^2)	10%	25%	50%	
	Smoke[2,3]	10,000 ft^2 (930 m^2)	15,000 ft^2 (1400 m^2)	5 – 10%	5 – 10%	5 – 25%	
B	Fire	700 ft^2 (65 m^2)	1000 ft^2 (93 m^2)	50%	100%	100%	1 – 3 Weeks
	Water	1500 ft^2 (140 m^2)	2000 ft^2 (185 m^2)	25%	50%	100%	
	Smoke[2,3]	10,000 ft^2 (930 m^2)	15,000 ft^2 (1400 m^2)	5 – 20%	5 - 30%	100%	
C	Fire	1200 ft^2 (95 m^2)	1500 ft^2 (140 m^2)	25%	100%	100%	1 - 7 Days
	Water	2500 ft^2 (185 m^2)	3500 ft^2 (325 m^2)	25%	25%	50%	
	Smoke[2,3]	20,000 ft^2 (1860 m^2)	30,000 ft^2 (2787 m^2)	5 – 10%	5 – 10%	5 – 25%	
D	Fire	1200 ft^2 (95 m^2)	1500 ft^2 (140 m^2)	50%	100%	100%	2 – 4 Weeks
	Water	2500 ft^2 (185 m^2)	3500 ft^2 (325 m^2)	40%	50%	100%	
	Smoke[2,3]	20,000 ft^2 (1860 m^2)	30,000 ft^2 (2787 m^2)	5 - 20%	5 - 30%	100%	
E	Fire	Sprinkler system design area		25%	100%	100%	8 – 16 Weeks
	Water	2 times Fire Area		25%	25%	50%	
	Smoke[2,3]	Entire Compartment/Room		5 - 10%	5-10%	5-25%	
F	Fire	Sprinkler system design area		50%	100%	100%	4 – 12 Weeks
	Water	2 times Fire Area		25%	50%	100%	
	Smoke[2,3]	Entire Compartment/Room		5 - 20%	5 - 30%	100%	
G	Fire	Sprinkler system design area		100%	100%	100%	4 – 12 Weeks
	Water	5 times Fire Area		25%	25%	50%	
	Smoke[2,3]	Entire Compartment/Room		5 - 20%	5-20%	25-100%[5]	
H	Fire	500 ft^2 (45 m^2)		25%	100%	100%	1 - 7 Days
	Water	2000 ft^2 (190 m^2)		25%	25%	100%	
	Smoke[2,3]	10,000 ft^2 (930 m^2)		5 – 10%	5-10%	5-25%	
I	Fire	500 ft^2 (45 m^2)		50%	100%	100%	2 – 4 Weeks
	Water	2000 ft^2 (190 m^2)		40%	40%	100%	
	Smoke[2,3]	10,000 ft^2 (930 m^2)		5-20%	10-20%	100%	
J	Fire	700 ft^2 (45 m^2)	1500 ft^2 (140 m^2)	25%	100%	100%	1 - 7 Days
	Water	2000 ft^2 (190 m^2)	4000 ft^2 (372 m^2)	25%	25%	100%	
	Smoke[2,3]	10,000 ft^2 (930 m^2)	25,000 ft^2 (2323 m^2)	5 – 10%	5-10%	5-25%	
K	Fire	700 ft^2 (45 m^2)	1500 ft^2 (140 m^2)	50%	100%	100%	2 – 4 Weeks
	Water	2000 ft^2 (190 m^2)	4000 ft^2 (372 m^2)	40%	40%	100%	
	Smoke[2,3]	10,000 ft^2 (930 m^2)	25,000 ft^2 (2323 m^2)	5-20%	10-20%	100%	

Note: Damage percentages apply only to the area involved, i.e., they are not percent of total building values.

1. Area involved is limited by the size for the compartment in which the fire originates. Any substantial partition (without substantial openings) will confine fire and smoke with effective sprinklers present. Water damage to floor areas may extend through/under doors and other similar penetrations.

   Damage areas are cumulative, i.e., fire damage areas are separate from water damage areas, which are separate from smoke damage areas.

   Example: Building Value = $10,000,000; Equipment Value = $25,000,000; WIP = $2,000,000

   Building Area = 100,000 ft^2; Scenario B (Highly Sensitive Occupancy)

   Building Damage = $10,000,000 / 100,000 ft^2 X [(50% X 700 ft^2) + (25% X 1500 ft^2) + (20% X 10,000 ft^2)]
   Equipment Damage = $25,000,000 / 100,000 ft^2 X [(100% X 700 ft^2) + (50% X 1500 ft^2) + (30% X 10,000 ft^2)]
   WIP Damage = $2,000,000 / 100,000 ft^2 X [(100% X 700 ft^2) + (100% X 1500 ft^2) + (100% X 10,000 ft^2)]

PD Total = $ 273,000 + $1,113,000 + $244,000 = $1,630,000

2. For large, open buildings such as warehouses, the percentage of the area affected by smoke can be estimated at:
   - 20 – 40% for buildings of about 50,000 ft^2 (4650 m^2)
   - 5 – 15% for buildings of about 200,000 ft^2 (18,600 m^2)
   - 1 – 2% for buildings of 800,000 ft^2 (74,300 m^2) or more

3. Use higher smoke damage percentages when area subject to smoke damage is limited by smaller compartment size.

4. For areas with pharmaceuticals, medical products, or food, all inventory/WIP in the space/room in which the fire occurs should be considered lost beyond recovery. All product that could possibly be exposed to smoke will be discarded due to liability concerns and should be considered 100% damaged.

5. The smoke damage is estimated at a maximum of 100% but can be lowered depending on commodity classification and sensitivity.

Table 1B – Typical APL Loss Estimate Damage – Sprinklers with Deficiencies

Deficiencies in more than one bolded category below then use:	APL = PML
Inadequate number of heads in design area or inadequate design pressure:	
a. Design area or pressure is ≥ 80% of required	Scenario G
b. Design area or pressure is < 80% of required	APL = PML
K=14 Heads in >35 ft (10.7 m) high buildings (ESFR) without In-Racks:	
a. Top of Storage to Ceiling Clearance ≤ 10 ft (3.0 m) OR Storage height ≤ 15 ft (4.6 m)	Scenario H[3] or Scenario I[4]
b. Clearance is greater than 10 ft (3.0 m) and storage is > 15 ft. (4.6 m)	APL = PML
c. Building height exceeds 40 ft (13.7 m)	APL = PML
Open-top containers (that don't drain immediately) are used:	
a. Containers on bottom tier only	Review all other deficiencies.[1]
b. Containers on top tier cover > 20% (or 20 ft^2) of each rack footprint	APL = PML
c. Containers on top two tiers cover > 20% of each rack footprint and rack is ≥ 3 tiers	APL = PML
d. Container density is less than b or c with adequate water supply	Scenario G
e. Container density is less than b or c with inadequate or tolerable water supply	APL = PML
Inadequate flue spaces (CMDA or CMSA):	
a. Inadequate transverse <u>or</u> longitudinal flue spaces	Scenario G
b. Inadequate transverse <u>and</u> longitudinal flue spaces	APL = PML
Inadequate flue spaces (ESFR):	
a. Inadequate transverse or longitudinal flue spaces	APL = PML
Smoke/Heat Vents (Warehousing):	
a. Smoke/Heat Vents provided in automatic mode [5] installed in accordance with NFPA 13 and NFPA 204.	Scenario G Review all other deficiencies.[1]
b. Smoke/Heat Vents provided in automatic mode [5] not installed in accordance with NFPA 13 and NFPA 204.	APL = PML
Obstructions (ESFR):	
a. Only one ESFR sprinkler head obstructed in the design area [2]	Scenario C [3] or Scenario D [4]
b. A maximum of three ESFR sprinkler heads obstructed in the design area [2]	Scenario G
c. Greater than three ESFR heads obstructed in the design area [2]	APL = PML
Obstructions (CMDA/CMSA):	
a. A maximum of three CMDA sprinkler heads obstructed in the design area [2]	Scenario E [3] or Scenario F [4]
b. A maximum of six CMDA sprinkler heads obstructed in the design area [2]	Scenario G
c. Greater than six CMDA heads obstructed in the design area [2]	APL = PML

Note:

1: This note is provided to instruct field engineers on the correct storage arrangement of open-top containers. Other deficiencies should be reviewed and if none identified, the APL must be completed using Figure 1A and using the decision “AS is reliable and adequate as installed”.

2: ESFR obstructions must be reviewed in accordance with NFPA 13, Chapter 14. Any obstruction to the discharge pattern development or to the ESFR sprinkler’s ability for water to reach the seat of a fire must be considered.

3: Scenario must be chosen for all non-sensitive occupancies.

4: Scenario must be chosen for all sensitive occupancies.
Sensitive occupancies would include but are not limited to the following – Manufacturing of Consumer Goods (Food, Beverage, Clothing), Sensitive Electronics, Pharmaceutical, and Medical Devices/Equipment.

PROBABLE MAXIMUM LOSS (PML)

The definition of the PML is the maximum likely loss that would result with a single protection system out of service at the time of the incident. Other active and passive protection systems are assumed to function as installed, employees will respond in accordance with procedures/training, and outside agencies will respond per standard procedures. This definition is also included in the standard report template. This definition implies that 95% or more of losses will be less than this estimate.

A *single protection system* is defined as all the fire protection provisions that will be disabled by the failure (or manual shut-off) of a single device or protection system component. The only exception is that an underground main failure should only be used as the PML condition if there is a history of underground main leaks/breaks at the site.

Many single device failures (or closures) result in loss of multiple systems. Where all sprinkler risers for a building are on a manifold that is controlled by a single control valve, the PML could be the loss of all sprinkler protection to the building. Where a single fire pump is required for an adequate water supply, and is installed and maintained in accordance with an approved standard, an impairment to the fire pump is **NOT** considered a credible PML scenario. For an extraction plant with a single foam concentrate pump or a single foam tank, failure/closure of the foam valve or the concentrate pump would likely be the PML scenario.

The PML system of failure need not be a sprinkler system. It could be a clean agent system for a tool in a semiconductor fab cleanroom. It could be an aspirating fire detection system in a data center. This Fire Practice addresses loss estimates for sprinkler systems failures only. Reference FP5601 for calculating loss estimates for electronic data processing equipment areas.

Complete failure of a combustion safeguard system is not considered to be a PML scenario. Unless there is a power outage, all aspects of a combustion safeguard can't fail together. The loss of any single function of that system does not usually result in an explosion since all other components working with it will also prevent explosive conditions. Loss of flame does not cause an explosion unless you also lose combustion air or induced draft air.

PML loss estimates should not normally be used to evaluate the cost effectiveness of recommendations. However, they are often considered practical worst case scenarios. As such they may be used in developing and evaluating emergency response and business continuity/disaster recovery plans. If the plan can deal with the PML scenario, then it can likely handle the more probable, lesser impact events.

Rules of Thumb for PML Loss Estimates:
1. Apply MFL loss estimate damage percentages (Table 4) to the PML fire area of the site.
2. Significantly delayed fire department notification and/or response (>40 minutes until arrival) will result in PML = MFL. See Table 3a or Table 3b for separation requirements.
3. Fire departments will not conduct aggressive interior attack when a fire is well established, and personnel/occupants are not in danger. Fire departments should only be credited for stopping well established fires where there is adequate physical or fire rated separation. Continue reading below.
4. When combustible construction is present, PML = MFL for the involved combustible structure(s). See Table 3a or Table 3b for separation requirements.

a. For exposed non-combustible or fire resistive construction, see below occupancy hazard specific guidance.

5. For noncombustible or fire resistive construction, a fully paid and well equipped fire department that arrives on site relatively promptly can be credited to stop a fire at any building separation greater than 150% of the building height or one-half (1/2) the distance listed in Table 3a or Table 3b (whichever separation distance is larger), or a NFPA 221/FMDS 1-21 qualifying fire wall rated at 2 hours or greater (without large, unprotected openings).
6. For extra hazard and storage occupancies without significant flammable/combustible liquids, with noncombustible/fire resistive construction, and a fully paid and well-equipped fire department the fire area may **only** be limited per item #5 above.
7. For light and ordinary hazard occupancies without flammable/combustible liquids and with adequately designed (per NFPA or FM) sprinkler systems, multiple sprinkler systems in the building of origin, central station supervision or 24 hour occupancy, and a well trained and equipped, quick responding fire department, the fire area may be limited to the size of the sprinkler system where fire originated plus 50%. Fire area may also be limited per item #5 above.
8. Facilities with significant flammable and combustible liquids need to be evaluated based on the quantity of liquid present, containment provisions, and other fire protection systems that can be used to minimize fire spread. See FP5300 (in development).

Special Considerations for High Rise Buildings:

The maximum effective reach of most fire departments' equipment for exterior firefighting is between 60 and 100 ft. (18 – 30 m) above grade. Therefore, exterior firefighting will be ineffective in buildings above that height and fire could spread externally above those elevations in certain conditions. Normally, the PML fire area for a high rise building will be one floor. However, certain conditions could allow the fire to spread further. Use the flow chart of Figure 2, below to determine the number of fire floors and Table 2 to determine the damage percentages.

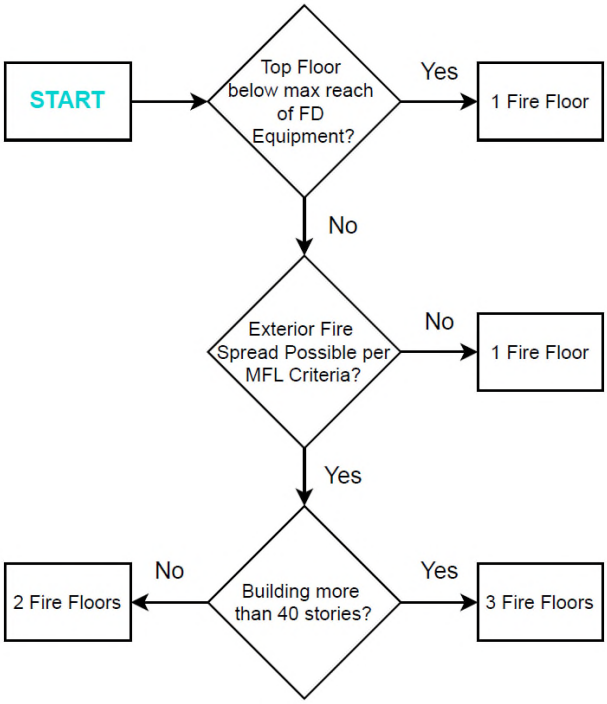


Figure 2 – High Rise PML Scenarios

Table 2 – PML High Rise Damage

	Damage		
Fire Floors Involved	***Fire***	***Smoke***	***Water***
1	*100% (average structure and contents value per floor)*	*20% (contents only) to 2 floors above*	*15% (contents only) to 2 floors below*
2		*20% (contents only) to 3 floors above*	*15% (contents only) to 5 floors below*
3		*20% (contents only) to 4 floors above*	*15% (contents only) to 6 floors below*

MAXIMUM FORESEEABLE LOSS (MFL)

The MFL is the loss that is likely with the failure of all active, nonredundant fire protection equipment and ineffective response from any personnel or outside agencies. Nearly 100 percent of all losses should be less than the MFL estimate. MFL estimates are used primarily by insurance companies to establish capacity requirements/limits. They are also sometimes used in business continuation planning efforts as described above for the PML.

The MFL scenario does not take any external make up capacities in consideration. Especially external backup capacities might not be existing at all times.

Rules of Thumb for MFL Loss Estimates (non High Rise):

Determining MFL Area:

- Use Table 3a or Table 3b as a guideline for minimum MFL space separation for typical occupancies and construction. The tables assume no combustible storage or materials in the space between the structures, with the exception of grass and shrubbery (no large trees or vehicle parking).
- Four hour rated high challenge (MFL) fire walls, designed, installed and maintained in accordance with NFPA 221 or FMDS 1-42 **may** be used to separate fire areas within a single building. MFL type fire walls are **exceptionally** rare and require a thorough analysis. Consult with the account team and chief engineer prior to crediting such fire wall in an MFL loss estimate. Use FM Data Sheet 1-22, *Criteria for Maximum Foreseeable Loss Fire Walls and Space Separation* should be used when in doubt of the limits for fire spread.

Table 3a – Minimum MFL Space Separation (ft)

	Building of Fire Origination *Construction[1]* *NFPA Occupancy Classification*								
Construction of Exposed Structure	*Fire Resistive*			*Noncombustible*			*Combustible[2]*		
	Light	*Ordinary*	*Extra*	*Light*	*Ordinary*	*Extra*	*Light*	*Ordinary*	*Extra*
Fire Resistive	30	40	60	40	50	80	50	80	100
Noncombustible	40	60	90	50	80	100	80	100	130
Combustible	80	100	130	100	130	150	130	150	200

Table 3b – Minimum MFL Space Separation (meters)

	Building of Fire Origination *Construction[1]* *NFPA Occupancy Classification*								
Construction of Exposed Structure	*Fire Resistive*			*Noncombustible*			*Combustible[2]*		
	Light	*Ordinary*	*Extra*	*Light*	*Ordinary*	*Extra*	*Light*	*Ordinary*	*Extra*
Fire Resistive	9	12	18	12	15	24	15	24	30
Noncombustible	12	18	27	15	24	30	24	30	40
Combustible	24	30	40	30	40	46	40	46	61

Note:

1: If openings in the facing wall of the Building of Fire Origination are greater than 25% of the wall area, use *Noncombustible* figures for Fire Resistive buildings and *Combustible* figures for Noncombustible buildings.
2: If the Building of Fire Origination is more than two stories or 20 ft. (6 m) high and not Fire Resistive construction, increase required space separation by 10 ft. (3 m) for each story above two or each 10 ft. (3 m) in building height over 20 ft. (6 m) high.

Determining MFL Damage:

Table 4 can be used as a starting point for determining the extent of damage in the MFL fire area. Where special occupancies/equipment are present, the MFL may involve those items because the loss from a standard fire MFL may be significantly less than the "specialty" loss. The percent damage estimates in Table 4 should be modified as appropriate for specific site/facility conditions. For example, a facility with significantly less combustible loading than is typical for the occupancy would likely have reduced building and content damage in an MFL fire. Conversely, unfavorable factors can increase the potential damage. Unfavorable factors include:

- Combustible loading significantly greater than typical for the occupancy
- Poor housekeeping
- Substandard maintenance of buildings and equipment
- Equipment or contents especially susceptible to damage from heat, water, or smoke.

Table 5 is provided below and is considered to be a cut-off point where structures will be considered condemned and required to be razed prior to reconstruction efforts. Building damage above the percent listed, should be considered a 100% loss unless other factors are identified. Undamaged equipment and inventory within the structure can be relocated prior to demolition and would not be considered a total loss.

Table 4 – Typical MFL Loss Estimate Damage Percentages

		Percent Damage			BI[1]
Occupancy	**Building Construction**	**Building**	**Equipment**	**Inventory/WIP**	**(Months)**
Hospital/Medical (non High-Rise)	Fire Resistive	10	60	100	6
	Noncombustible	30	80	100	9
	Combustible	100	100	100	18
	Medical equipment and instruments have a long lead time. If only part of the equipment is damaged, it will affect the overall diagnostic capabilities of the hospital.				
Office (non High-Rise)	Fire Resistive	40	40	50	3 – 6
	Heavy Noncombustible	50	40	50	3 – 6
	Light Noncombustible	60	50	60	3 – 6
	Combustible	100	100	100	6 – 9
Hydroelectric Power Generation	Fire Resistive	5	20	10	12
	Heavy Noncombustible	5	20	20	12
	Light Noncombustible	30	20	20	12
	Combustible	100	100	100	18
	The MFL would consider full damage of the installed turbine and replacement of the turbine.				
Data Center, Telecommunications	Fire Resistive	10	30	30	3
	Heavy Noncombustible	10	30	30	3
	Light Noncombustible	30	30	30	6
	Combustible	100	100	100	18
	Fire Resistive	10	20	20	3

Machine Shop, Light Metal Fabrication (Limited Combustible Liquids)	Heavy Noncombustible	10	20	20	3
	Light Noncombustible	30	30	30	6
	Combustible	100	100	100	12
Mineral Products	Fire Resistive	10	20	10	6
	Heavy Noncombustible	10	20	20	6
	Light Noncombustible	30	30	30	9
	Combustible	100	100	50 – 100 [2]	12
Beverage Processing and Bottling	Fire Resistive	40	40	100	9
	Heavy Noncombustible	50	40	100	12
	Light Noncombustible	60	50	100	12
	Combustible	100	100	100	18
Retail (non “Big Box”)	Fire Resistive	50	60	100	9 – 12
	Heavy Noncombustible	70	70	100	9 – 12
	Light Noncombustible	80	70	100	9 – 12
	Combustible	100	100	100	9 – 12
Food Processing	Fire Resistive	50	50	100	9 – 12
	Heavy Noncombustible	60	60	100	12
	Light Noncombustible	70	60	100	12
	Combustible	100	100	100	12
Electronic/Electrical Equipment Manufacturing	Fire Resistive	50	60	100	9
	Heavy Noncombustible	70	70	100	12
	Light Noncombustible	80	70	100	18
	Combustible	100	100	100	18
Metal Smelting, Ore Refining, Foundry, Forge, Rolling Mill	Fire Resistive	50	60	40	18
	Heavy Noncombustible	70	70	50	18
	Light Noncombustible	80	70	80	18
	Combustible	100	80	80	18
Big Box Retail	Fire Resistive	50	60	60	6 – 9
	Heavy Noncombustible	70	80	80	9 – 12
	Light Noncombustible	90	80	90	9 – 12
	Combustible	100	100	100	9 – 12
Box Manufacturing, Printing	Fire Resistive	50	70	100	9 – 12
	Heavy Noncombustible	70	80	100	9 – 12
	Light Noncombustible	90	80	100	18
	Combustible	100	100	100	18
Vehicle Manufacturing	Fire Resistive	50	50	50	12
	Heavy Noncombustible	60	60	60	15
	Light Noncombustible	70	60	60	18
	Combustible	100	100	100	24
Brewery, Distillery, Winery	Fire Resistive	50	60	100	12
	Heavy Noncombustible	70	70	100	12
	Light Noncombustible	80	70	100	18
	Combustible	100	100	100	18
Wood and Plastic Products Mfg., Sawmill, Pulp Mill	Fire Resistive	70	90	100	18
	Heavy Noncombustible	80	90	100	18
	Light Noncombustible	90	90	100	18
	Combustible	100	100	100	18
Cotton Mill, Sugar Mill	Fire Resistive	80	100	100	18
	Heavy Noncombustible	90	100	100	18
	Light Noncombustible	100	100	100	18
	Combustible	100	100	100	18
Rubber Manufacturing	Fire Resistive	70	90	100	12

	Heavy Noncombustible	80	90	100	18
	Light Noncombustible	90	90	100	18
	Combustible	100	100	100	18
Warehousing (includes warehousing of combustible materials in other occupancies)	Fire Resistive	70	90	90	9
	Heavy Noncombustible	80	90	90	12
	Light Noncombustible	100	100	100	12
	Combustible	100	100	100	12
Warehousing (Refrigerated)	Fire Resistive	70	90	100	12
	Heavy Noncombustible	80	90	100	18
	Light Noncombustible	100	100	100	18
	Combustible	100	100	100	18
Aircraft Hangar	Fire Resistive	90	100	100	18
	Heavy Noncombustible	100	100	100	18
	Light Noncombustible	100	100	100	18
	Combustible	100	100	100	18
Perfumes, Cosmetics, Drugs, Pharmaceutical	Fire Resistive	80	100	100	24
	Heavy Noncombustible	90	100	100	>=24
	Light Noncombustible	100	100	100	>=24
	Combustible	100	100	100	>=24
Semiconductor Mfg.	Fire Resistive	20	90	100	>=24
	Heavy Noncombustible	20	90	100	>=24
	Light Noncombustible	30	90	100	>=24
	Combustible	100	100	100	>=24
Grain Elevator, Grain Milling	Fire Resistive	90	100	100	18
	Heavy Noncombustible	100	100	100	18
	Light Noncombustible	100	100	100	18
	Combustible	100	100	100	18
Hotel, Residential, Schools/Academic, Museum	Fire Resistive	50	50	70	9 – 12
	Heavy Noncombustible	60	60	70	9 – 12
	Light Noncombustible	70	60	80	9 – 12
	Combustible	100	100	100	9 – 12

Note:

1: BI will be dependent on replacement time of equipment as well as the structure. Qualifying processes can take 3 to 6 months on top of equipment lead times (pharmaceutical, semiconductor, food industry, etc.). This needs to be investigated while inquiring about BI issues. In addition, in some jurisdictions, permitting requirements may delay reconstruction – even when damage is in the 50% range (replacement must meet latest codes).

2: Coal Mineral Products will be a full loss of stock. All others are estimated at 50%.

Table 5 – Maximum % Building Damage that Defaults to Full Structure Loss

Area Ft² (m²)	***Light Non-Combustible***	***Heavy Non-Combustible***	***Fire Resistive***
<50,000 (4,650)	60%	60%	60%
50,000 – 200,000 (4,650 – 18,600)	65%	70%	70%
200,000 – 500,000 (18,600 – 46,450)	70%	75%	75%
500,000 – 1,000,000 (46,450 – 92,900)	80%	85%	85%
1,000,000+ (92,900+)	90%	90%	90%

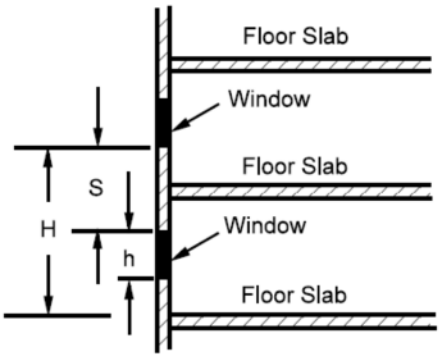
Note:

Oftentimes repair of a structure will be more costly than a full demolition and reconstruction. The above table should be considered when determining the total building loss that the facility will see. While the building damage may be limited to a lower percentage than 100%, the building will likely be condemned and restoration will require complete demolition. In these cases, equipment and/or stock may still be salvageable.

Determining MFL Area for High Rise Buildings:

Exterior fire spread in a high rise building is not normally considered possible per FM Data Sheet 1-3 when the distance from the floor slab of a floor to the bottom of the window glass of the floor above is at least 2.8 times the height of the window. If windows can be opened by occupants, the ratio should be 3.8 times the height of the window.

$H \geq 2.8h \rightarrow$ No exterior spread (closed windows)
$H \geq 3.8h \rightarrow$ No exterior spread (open windows)



FMDS 1-3, Figure 1A (Partial)

Note that a blank exterior wall mechanical floor can serve as a vertical fire stop since the window height is zero.

Where combustible construction materials are used on the exterior of the building (e.g. EIFS – foam insulation covered with plaster/stucco, often used in very hot or very cold climates; or insulated metal panels; or other combustible plastic construction materials), external fire spread should be assumed for all floors where these combustible materials are used.

Interior fire spread is limited by good sealing of vertical openings.

- All vertical openings (stairwells, shafts, pipe/electrical chases, laundry/mail chutes, etc.) must have automatically closing doors with fire ratings of at least 1.5 hours. A plain metal panel does not achieve this objective.
- Electrical conduit/cable tray and pipe penetrations through floors (outside of rated chases) and through walls into rated chases must be sealed with material/fire stopping that provides at least a 2 hour fire resistance rating.
- The space between the edge of the floor slab and the exterior curtain wall must be sealed with noncombustible material designed for that purpose. Usually, mineral wool or ceramic fiber is used for this purpose. Standard fiberglass insulation is not normally acceptable for this purpose. This material must be held securely in place by noncombustible retainers.

MFL Scenario – no exterior or interior spread potential: Fire starts on the floor below the floor of greatest value. Fire progresses to floor above through stair or other propped open penetration. Damage to two fire floors is 100% of average structure value per square foot for the floors involved plus 100% damage to contents. Smoke damage to four floors above is 20% of content values and water damage for six floors below is 15% of content values. BI is normally 9 – 12 months.

MFL Scenario – exterior or interior fire spread potential: Fire starts on ground floor and progress up through the structure to a fire stop mechanical floor or to the top of the building. Alternatively, fire starts above the top fire stop mechanical floor and progresses to the top of the building. Structure and content damage is 100% to the floors involved. Smoke damage is 20% to six floors above fire area (if any) and water damage is 15% to the 10 floors below the fire area (if any). BI is 18 to 24 months.

Revision Record

Rev.	Description		
Orig	Original Issue		
A	• Complete re-write adding flow charts, scenarios, damage percentage tables.		

B	General update.
	• Split Figure 1 and Table 1 in two
	  o Separate storage from other occupancies
	  o Provide guidance for minor ESFR/CMSA deficiencies
	• Add Data Center and office scenarios		

C	• Fix typos and clarify Building Damage Area for some scenarios		
D	• Revised loss scenarios for pharmaceutical and food storage facilities.		
D.1	Fixed typo in Table 1B • Added definition of *single protection system* for PML definition.		
D.2	• Revised Table 1B for K=14 ESFR in 40 ft. high buildings		
D.3	• Updated logo only		
E	Complete major overhaul including: • Rewrite on all existing tables; o Removed part of Table 1B for additional loss scenarios related to ESFR/CMSA and pointed back to main scenario table. • Rewrite on all existing figures; • Purpose clarified to be for fire only hazards (including external fires) • General and APL sections rewrote to include guidance for certain special hazards • Updated PML rule of thumbs • Added Table 5 for building % damage cut-off before complete loss • Split Figure 1B into Figure 1B / 1C to differentiate between sensitive/non-sensitive non-storage occupancies • Removed write-ups of occupancy specific APLs and pointed to specific Fire practices. Preparedness plan information updated to be consistent with Windstorm Guidelines document. Reference to that document added. • Replace Fujita Scale table with Enhance Fujita Scale • Update of MFL BI	DB, ML	JP